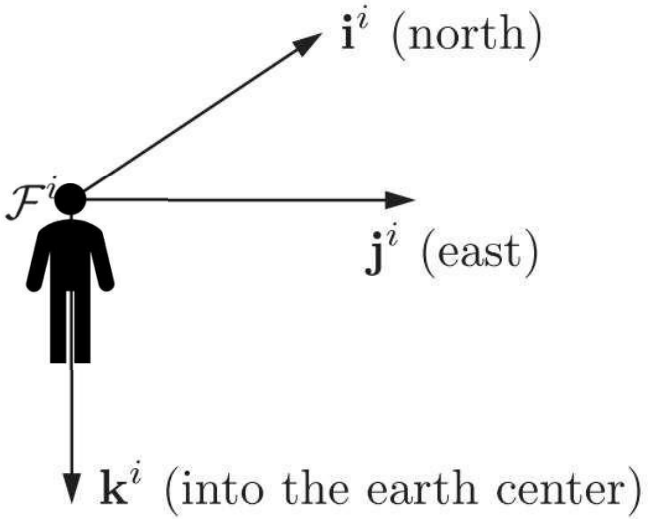
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Tutorial 2 :Kinematics and Dynamics



Kinematics-Dynamics Tutorial



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Conventions and tutorial tasks

Body axes: x forward, y right, z down.

NED position: north, east and down. Use $g = 9.81\ m/s^2$.

F_x, F_y and F_z are total external forces in body axes, including gravity.

Rigid body, constant mass and diagonal inertia about the centre of mass.

Tasks: vertical motion, rotating-frame velocity, rotational motion, then the system block diagram.

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Reference: translational and rotational dynamics

Translational dynamics

$$\dot{u} = rv - qw + \frac{F_x}{m}$$

$$\dot{v} = pw - ru + \frac{F_y}{m}$$

$$\dot{w} = qu - pv + \frac{F_z}{m}$$

Rotational dynamics (diagonal inertia)

$$\dot{p} = \frac{M_x + (J_y - J_z)qr}{J_x}$$

$$\dot{q} = \frac{M_y + (J_z - J_x)pr}{J_y}$$

$$\dot{r} = \frac{M_z + (J_x - J_y)pq}{J_z}$$

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Reference: position and attitude kinematics

Position rate: rotate velocity components into NED.

$$\begin{bmatrix} \dot{p}_N \\ \dot{p}_E \\ \dot{p}_D \end{bmatrix} = R_b^n \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

Euler-angle rates for the ZYX convention

$$\dot{\varphi} = p + qsin\varphi tan\theta + rcos\varphi tan\theta$$

$$\theta = qcos\varphi - rsin\varphi$$

$$\dot{\psi} = \frac{qsin\varphi + rcos\varphi}{cos\theta}$$

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Question 1: Vertical motion and hover

A level UAV has mass 1.60 kg and upward thrust 18.0 N. It moves vertically with no body rotation. Neglect drag.

Use z positive downwards and let w be the vertical velocity.

(a) Write the two state equations for z and w.

(b) Calculate the vertical acceleration and state its direction.

(c) If w = +0.800 m/s, is the descent speeding up or slowing down?

(d) State both conditions required for sustained hover.

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Question 1: Solution

$$\dot{z} = w, \dot{w} = g - \frac{T}{m}$$

$$\dot{w} = 9.81 - \frac{18.0}{1.60} = -1.44\ m/s^2$$

Acceleration is upwards because downwards is positive.

With w = +0.800 m/s, the UAV is descending but slowing down.

$$Hover\colon\ w = 0, T = mg = 15.696\ N$$

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Question 2: Velocity in rotating body axes

At one instant, m = 2.50 kg. All vectors use body axes.

$$v_b = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} m/s, \omega_b = \begin{bmatrix} 0 \\ 0 \\ 0.400 \end{bmatrix} rad/s$$

$$F^b = \begin{bmatrix} 5.00 \\ 0 \\ 0 \end{bmatrix} N$$

(a) Calculate $\dot{u}$, $\dot{v}$ and $\dot{w}$.

(b) Explain why $\dot{v}$ can be nonzero when $F_y = 0$.

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Question 2: Solution

$$\dot{u} = 0.400(1) + \frac{5.00}{2.50} = 2.40\ m/s^2$$

$$\dot{v} = -0.400(3) = -1.20\ m/s^2, \dot{w} = 0$$

The body axes rotate, so the velocity components change even without a lateral force.

$$a_b = \dot{v}_b + \omega_b \times v_b$$

$$\omega_b \times v_b = [-0.400, 1.20, 0]^T\ m/s^2$$

$$a_b = [2.00, 0, 0]^T\ m/s^2$$

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Question 3: Rotational dynamics

A rigid UAV has diagonal inertia:

$$J_x = 0.0250, J_y = 0.0350, J_z = 0.0450\ kg\ m^2$$

$$M_x = 0.0800\ N\ m, M_y = M_z = 0$$

(a) At an instant with p = q = r = 0, calculate $\dot{p}$.

(b) In a separate case, p = 0, q = 1.50 rad/s and r = 2.00 rad/s. Calculate $\dot{p}$, $\dot{q}$ and $\dot{r}$.

(c) Explain why $\dot{p}$ differs between the two cases.

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Question 3: Solution

$$(a)\ \dot{p} = \frac{0.0800}{0.0250} = 3.20\ rad/s^2$$

$$(b)\ (J_y - J_z)qr = (0.0350 - 0.0450)(1.50)(2.00)$$

$$= -0.0300\ N\ m$$

$$\dot{p} = \frac{0.0800 - 0.0300}{0.0250} = 2.00\ rad/s^2$$

$$\dot{q} = 0, \dot{r} = 0\ at\ this\ instant$$

Rotation about the other axes contributes a coupling term.

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Question 4: System model

Using the multirotor kinematics and dynamics equations, draw a block diagram of the rigid-body system.

(a) List the 12 states and group them into four vectors.

(b) Use total body forces and moments as the model inputs.

(c) Show the four equation blocks, integrators and the state signals each block requires.

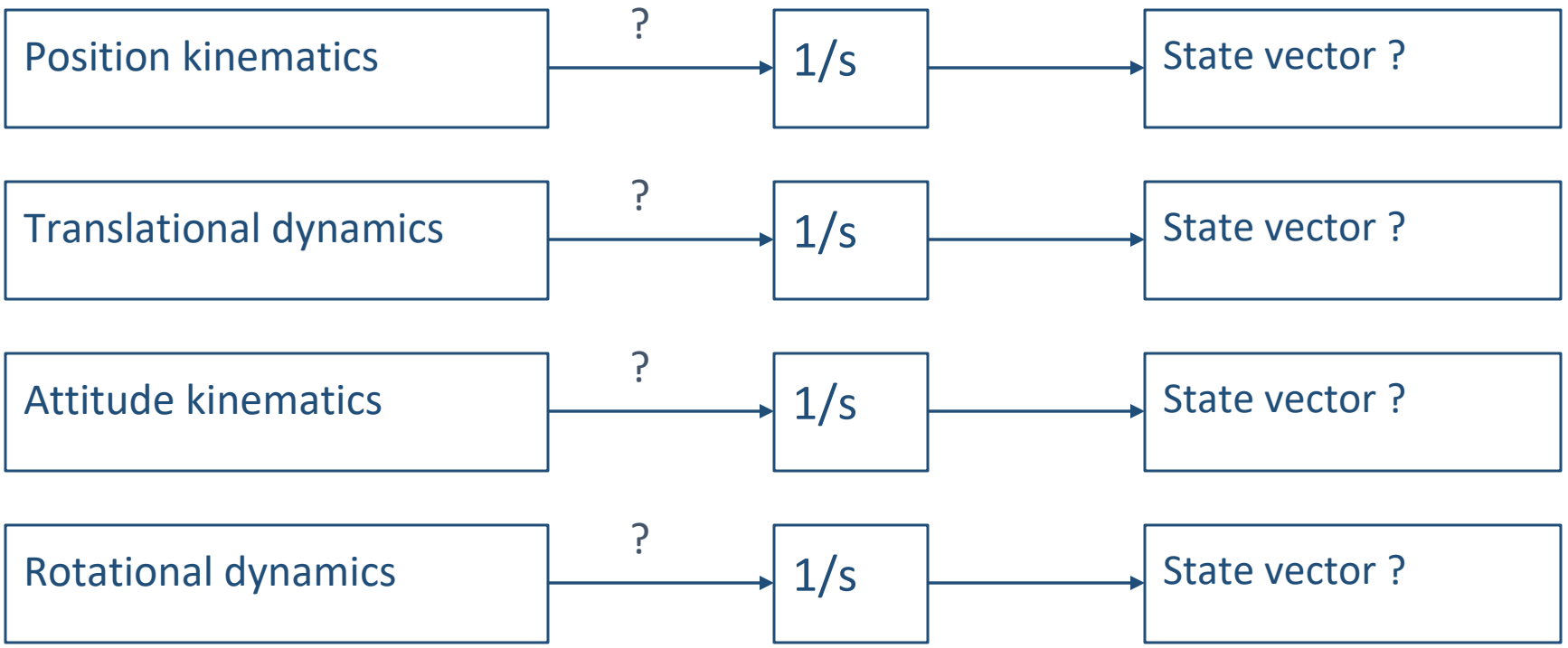
(d) Identify the initial conditions needed by the integrators.

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Question 4: Diagram scaffold

Complete the derivatives and states. Add the required input signals.



What initial value belongs to each integrator?

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Question 4: dynamics branches

Moments and angular velocity enter rotational dynamics.



Required signals: M_x, M_y, M_z and p, q, r. Parameter: J.

Forces, velocity and body rates enter translational dynamics.



Required signals: F_x, F_y, F_z, u, v, w and p, q, r. Parameter: m.

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Question 4: Kinematics branches

Body rates and attitude enter attitude kinematics.



Required signals: p, q, r, ϕ, ψ and θ.

Body velocity and attitude enter position kinematics.

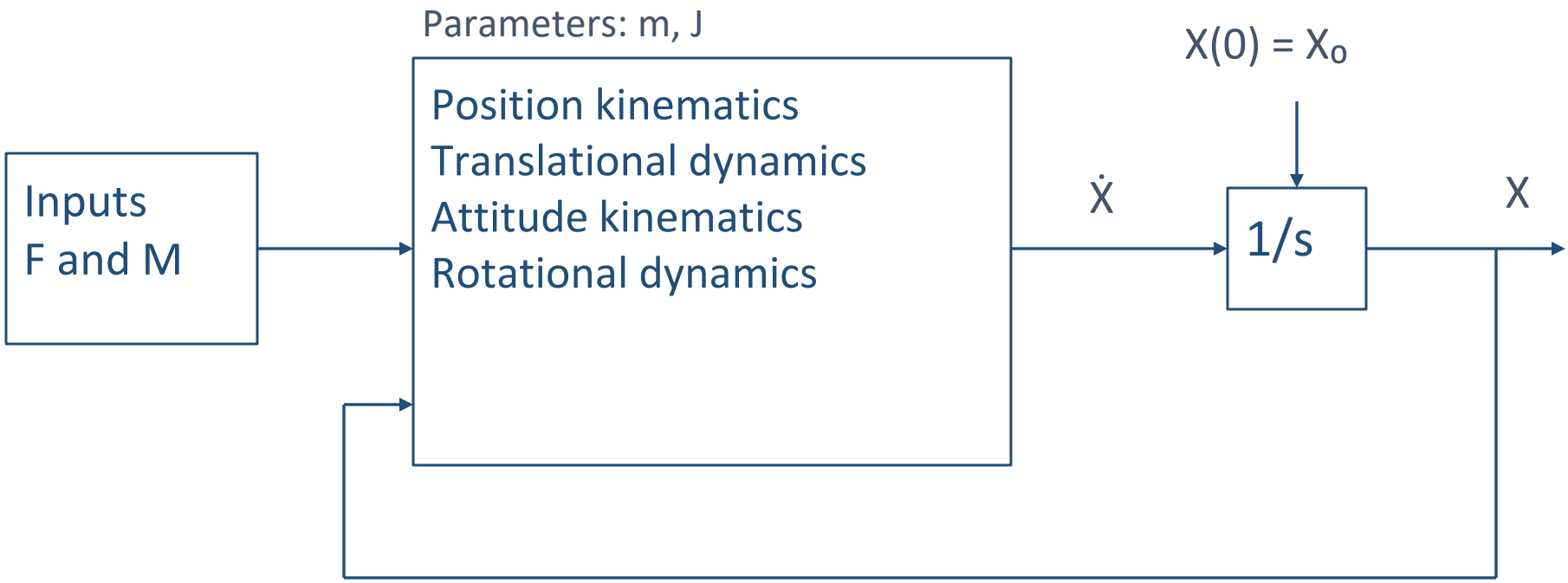


Required signals: u, v, w, ϕ, θ and ψ.

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Question 4: Complete state model

$X = [p_N, p_E, p_D, u, v, w, \phi, \theta, \psi, p, q, r]^T$



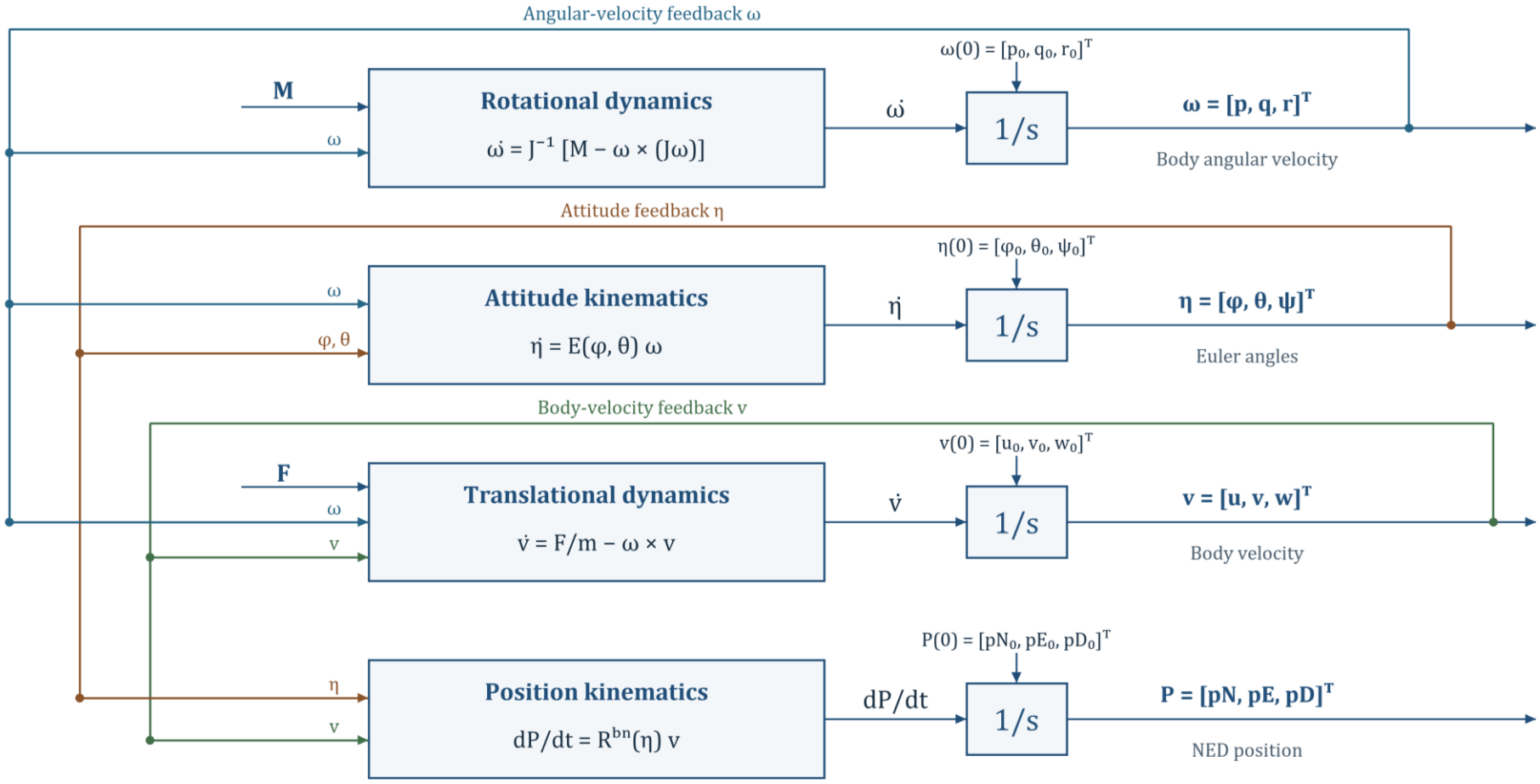
State feedback within the model

Each vector integrator needs its initial state values.

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Question 4: Complete state model



- **Connected junction | Crossing lines without a dot are not connected.**

Each 1/s integrates 3 states. F includes gravity; F, M, v and ω use body axes. Constant mass m and body-fixed inertia J.

E: Euler-rate mapping ($\theta \neq \pm 90°$). R^{bn}: body-to-NED rotation. Moments are about the centre of mass. P is position.

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Reference: body-to-NED rotation matrix

ZYX Euler angles. Cα = cos α and Sα = sin α.

$$R_b^n = \begin{bmatrix} C\theta C\psi & S\varphi S\theta C\psi - C\varphi S\psi & C\varphi S\theta C\psi + S\varphi S\psi \\ C\theta S\psi & S\varphi S\theta S\psi + C\varphi C\psi & C\varphi S\theta S\psi - S\varphi C\psi \\ -S\theta & S\varphi C\theta & C\varphi C\theta \end{bmatrix}$$

$$\varphi = \theta = \psi = 0 \Rightarrow R_b^n = I$$

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